

SWEEP rule on the KD-Toric code

February 9, 2026

1 SWEEP Rule Decoder on the Toric

The decoder works as follows: each vertex v looks for a suggestion $\hat{v}$ on $\uparrow(v) \cap lk_2$ that matches σ_v^+ . If there are more than one, then pick the one which has the minimal weight.

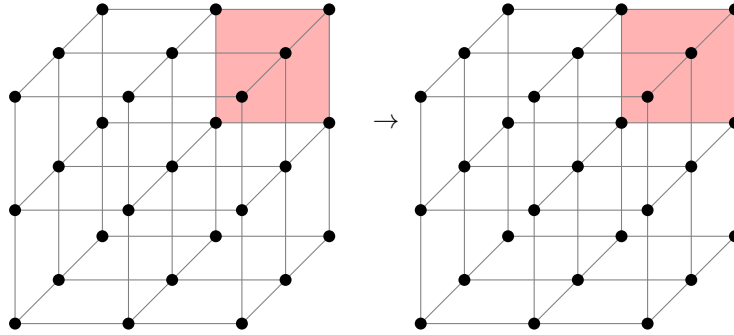


Figure 1: SWEEP rule

2 Span Expansion

Define the potentials functions $L_i : \mathbb{R}^d \rightarrow \mathbb{R}$ as follow¹:

$$L_i(\mathbf{x}) = \begin{cases} \mathbf{x}_i & i < d + 1 \\ -\langle \mathbf{1}, \mathbf{x} \rangle & i = d + 1 \end{cases}$$

The **Size** of a subset S is defined as:

$$\text{Size}(S) = \sum_i \max_{\mathbf{x} \in S} L_i(\mathbf{x})$$

Numerically, we show that the span property holds for the checks. Namely, let e be an unsatisfied check at time t , and let H_e be the unsatisfied checks at time $t - 1$. Then there is a set $H'_e \subset H_e$ such that:

1. For any $i < d + 1$ there exists $e' \in H'_e$ such $L_i(e') \geq L_i(e)$.
2. There exists $e' \in H'_e$ such that $L_{d+1}(e') \leq L_{d+1}(e) + 1$.

Proof. Consider a vertex v which at time t adjoins to an unsatisfied edge e , let u, w be vertices which have a positive faces that adjoin to e .

¹Notice that we use opposite signs relative to Berman and Peter Gacs, yet we agree with Perskil.

1. If $\hat{u}|_e \neq \emptyset$, then the parallel edge to e going out from u belongs to H_e . Denote it by e_u , taking e_u satisfies at least two (three in 4D) of the first d inequalities and the last $d + 1$ th inequality. Denote by $x \in [d]$ the index of the inequality which isn't satisfied by e_u . So it remains to prove the existence of $e' \in H_e$ for which $L_x(e') \geq L_x(e)$.

Consider the assignment of bits at time $t - 1$ over the complex, and consider the graph $G = (\mathcal{F}, E)$ induced by associating vertices with any face that is set to 1 by the assignment. Two vertices are connected if their origin faces share an edge. Let $K \subset \mathcal{F}$ be a connected component of faces that contains the faces supported on u .

We say that K is d far from u if:

$$\max_{f \in K} |f_x - u_x| \leq d$$

Now, we will prove by induction on d that there is a face $f \in K$ with $f_x \geq v_x$ such one of its edge is not satisfied.

- (a) **Base.** The edge from v which point at the $\hat{z}$ direction must not be satisfied.

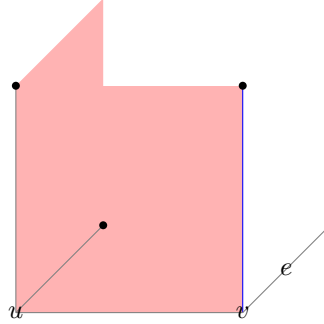


Figure 2: The induction base: In red, we have the faces on the support of u . The blue edge has the same x -coordinate as e . Any attempt to zero out the syndrome on the edge results in adding a face that is at least 2-far from u . Any configuration which is at most 1-far from u has to have that local view, yet might be connected to more faces which don't adjoin the blue edge.

- (b) **Assumption.** Assume that for any K which is at most d far from u there is a face, with adjoins edge that is both has a x -coordinate that is greater than u_x and is not satisfied.
 - (c) **Induction Step.** Notice that if by substitute a stabilizer from K we get G' that is at most d' far from u for some $d' < d$ then we done.
2. Else, if for any u it holds that $\hat{u}|_e = \emptyset$, then it must be that $e \in H_e$ because $\hat{v}$ can only decrease the syndrome on v . Moreover, it follows that $\hat{v} = \emptyset$ (otherwise, the suggestion would zero out the syndrome on $\uparrow(v) \cap lk_2$). Thus, $\sigma_v \not\subset \uparrow(v) \cap lk_2 \Rightarrow$ there is also an edge $e' \in H_e$ where $e' \in \downarrow(v)$. In that case, take $e \in H_e$ as the edge satisfies the first d equalities and e' for satisfying the $d + 1$ th inequality.

□